

0.0.1 Interacting Multipoles of Rank 1 and 1

$$-1 \cdot T_{\alpha\beta} \mu_{\alpha}^B \mu_{\beta}^A = R_z^{-5} \cdot -1 \cdot \begin{bmatrix} +3 \cdot R_{\alpha} \cdot R_{\beta} \cdot \mu_{\alpha}^B \cdot \mu_{\beta}^A \\ -1 \cdot R_z^2 \cdot \delta(\alpha, \beta) \cdot \mu_{\alpha}^B \cdot \mu_{\beta}^A \end{bmatrix} \quad (1)$$

Simplify deltas:

$$-1 \cdot T_{\alpha\beta} \mu_{\alpha}^B \mu_{\beta}^A = R_z^{-5} \cdot -1 \cdot \begin{bmatrix} +3 \cdot R_{\alpha} \cdot R_{\beta} \cdot \mu_{\alpha}^B \cdot \mu_{\beta}^A \\ -1 \cdot R_z^2 \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^B \cdot \mu_{\alpha}^A \end{bmatrix} \quad (2a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-5} \cdot -1 \cdot \begin{bmatrix} +3 \cdot R_{\alpha} \cdot R_{\beta} \cdot \mu_{\alpha}^B \cdot \mu_{\beta}^A \\ -1 \cdot R_z^2 \cdot \mu_{\alpha}^B \cdot \mu_{\alpha}^A \end{bmatrix} \quad (2b)$$

Simplify R:

$$-1 \cdot T_{\alpha\beta} \mu_{\alpha}^B \mu_{\beta}^A = R_z^{-5} \cdot -1 \cdot \begin{bmatrix} +3 \cdot R_z \cdot R_z \cdot \mu_z^B \cdot \mu_z^A \\ -1 \cdot R_z^2 \cdot \mu_{\alpha}^B \cdot \mu_{\alpha}^A \end{bmatrix} \quad (3a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-5} \cdot -1 \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_{\alpha}^B \cdot \mu_{\alpha}^A \end{bmatrix} \quad (3b)$$

Multiply Out Sum:

$$-1 \cdot T_{\alpha\beta} \mu_{\alpha}^B \mu_{\beta}^A = R_z^{-5} \cdot -1 \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_x^B \cdot \mu_x^A \\ -1 \cdot R_z^2 \cdot \mu_y^B \cdot \mu_y^A \\ -1 \cdot R_z^2 \cdot \mu_z^B \cdot \mu_z^A \end{bmatrix} \quad (4a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-5} \cdot -1 \cdot \begin{bmatrix} -1 \cdot R_z^2 \cdot \mu_y^B \cdot \mu_y^A \end{bmatrix} \quad (4b)$$

Combining everything:

$$-1 \cdot T_{\alpha\beta} \mu_{\alpha}^B \mu_{\beta}^A = \begin{bmatrix} +1 \cdot R_z^{-3} \cdot \mu_y^B \cdot \mu_y^A \end{bmatrix} \quad (5)$$